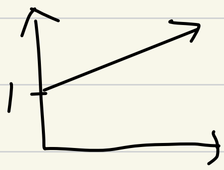
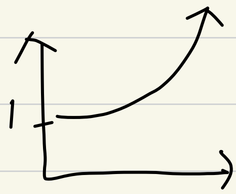
Chapter 1

Accumulated Value (AV) or
Future Value (FV)

accumulate function $a(t)$



simple interest



Compound Interest.

amount function $A_K(t)$:
denotes accumulated value at time t
of \$K invested at time 0.

$\rightarrow A_K(t) = K \cdot a(t)$ (assumption)

Effective Interest Rate on $[t_1, t_2]$

$$\rightarrow i[t_1, t_2] = \frac{A_K(t_2) - A_K(t_1)}{A_K(t_1)}$$

$$= \frac{\text{interest earned on } [t_1, t_2]}{\text{balance at the beginning of the period.}}$$

$\rightarrow$ assume $A_K(t) = K \cdot a(t)$

$$i[t_1, t_2] = \frac{a(t_2) - a(t_1)}{a(t_1)}$$

and for $\mathbb{Z}^+ n$ $\quad i[n-1, n] = \dfrac{a(n) - a(n-1)}{a(n-1)} := i_n$

Interest rate in nth period.

Simple Interest

constant

An investment accumulates with simple interest of i per year $\Leftrightarrow$ $a(t) = 1 + it$, $t \geq 0$ (years)

$$\text{(EIR)}\quad i_n = \frac{a(n) - a(n-1)}{a(n-1)} = \frac{i}{1 + i(n-1)}$$

$\hookrightarrow \to 0$ as $n \to \infty$

measure of time t:

(a) Exact simple interest method (actual / actual)
→ t = # days for loan / # days a year (365)

(b) Ordinary simple interest method / Banker's Rule
→ t = # days for loan / 365
assumes 365 days a year

(c) 30/360 Rule
→ Suppose (y_1, m_1, d_1) loan start and (y_2, m_2, d_2) loan end.
→ $t = D/360$
, where $D = 360(y_2 - y_1) + 30(m_2 - m_1) + (d_2 - d_1)$
→ assumes 360 and 30

Geometric Series:

$$a + ar + ar^2 + \dots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$$

Compound Interest. most important.

An investment accumulates with compound interest of i per year $\Leftrightarrow$ $a(t) = (1+i)^t$, $t \geq 0$

$\rightarrow$ $a(t+s) = a(t) \cdot a(s)$, $s, t \geq 0$

(EIR) $i_n = \dfrac{a(n) - a(n-1)}{a(n-1)} = i$ constant.

Present Value (PV)

$\$C$ at time t $\Rightarrow$ $C = PV \cdot a(t)$

$\Rightarrow PV = C \cdot \dfrac{1}{a(t)}$

Discount function $v(t) = \dfrac{1}{a(t)}$

denotes PV at time 0 of 1\$ due at t

a.k.a normally $a(t) = \$1 \rightarrow \$$ sth

$\$$ sth $\rightarrow \$1$

$= v(t)$

under compound interest

, $v(t) = \dfrac{1}{a(t)} = \dfrac{1}{(1+i)^t} = \left(\dfrac{1}{1+i}\right)^t = v^t$

where $v := \dfrac{1}{1+i}$ aka discount factor per year.

(constant)

Effective discount Rate $[t_1, t_2]$

$$d_{[t_1,t_2]} = \frac{A_K(t_2) - A_K(t_1)}{A_K(t_2)}$$

$$= \frac{\text{discount amount on } [t_1, t_2]}{\text{balance at the end of period.}}$$

An investment accumulates simple discount rate d per year $\Leftrightarrow$ $a(t) = \frac{1}{1-dt}$

and, $v(t) = 1 - dt$

An investment accumulates compound discount rate d (constant) per year $\Leftrightarrow$ $a(t) = \frac{1}{(1-d)^t}$

and, $v(t) = (1-d)^t$

Compound interest r. = Compound discount r.

$$\Rightarrow \text{if } d = \frac{i}{1+i} \Leftrightarrow i = \frac{d}{1-d}$$

$$\Rightarrow V = 1 - d.$$

(D factor)

Nominal rate of interest accumulates

m times per year of $i^{(m)}$

$$\Leftrightarrow a(t) = \left(1 + \frac{i^{(m)}}{m}\right)^{mt}, \quad t \geq 0$$

(compound since $a(t+s) = a(t)\,a(s)$)

↳ 名目利率，利息≠購買力 (no inflation)

$i^{(m)} \Rightarrow$ compounded m times a year.

for $i = i^{(m)}$, $1+i = \left(1+\frac{i^{(m)}}{m}\right)^m$ $\quad i^{(1)} = i$

cause compounded annually.

Nominal rate of discount

accumulates m times per year of $d^{(m)}$

$$\Leftrightarrow \quad a(t) = \frac{1}{\left(1-\frac{d^{(m)}}{m}\right)^{mt}}, \quad t \geq 0$$

compound since $a(t+s) = a(t)a(s)$

for $d = d^{(m)}$, $1-d = \left(1-\frac{d^{(m)}}{m}\right)^m$ $\quad 1+i = \frac{1}{1-d}$

$* \; i > i^{(m)} > d^{(m)} > d$

$i^{(m)}$ decrease as m increase.

Force of interest

$$\delta_t = \frac{d}{dt}\ln a(t) = \frac{1}{a(t)} \cdot \frac{da(t)}{dt}$$

relative instantaneous r.o.c at t

$$\Rightarrow a(t) = e^{\int_0^t \delta_r \, dt}$$

$$\Rightarrow v(t) = e^{-\int_0^t \delta \, dr}$$

since $v(t) = \frac{1}{a(t)}$

$\delta_t = \ln(1+\bar{i})$ under compound interest $\bar{i}$

$a(t) = e^{\delta t}$

$v(t) = e^{-\delta t}$

Under Compound interest :

$\Rightarrow a(t) = e^{\delta t}$

$\Rightarrow v(t) = e^{-\delta t}$

and, $\lim_{m\to\infty} i^{(m)} = \lim_{m\to\infty} d^{(m)} = \delta$

$\bar{i} > i^{(m)} > \delta > d^{(m)} > d$

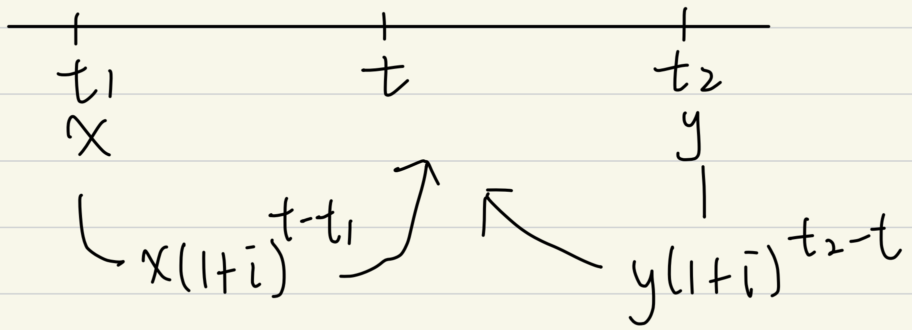
Chapter 2.

T-bill = US government bonds but < 365 days.

usually uses simple interest

time-t equation of value : (equivalence date)

sets value of two payments \$X & \$Y equal at a comparison date, t.



Satisfies time-t eqn. of value if

$$x(1+i)^{t-t_1} = y(1+i)^{t_2-t}$$

Property of transitivity :

X equivalent Z & Y equivalent Z
$\Rightarrow$ X equivalent Y

Equivalent payment streams :

at $t=0$, deposit must equivalent withdraws.

Inflation: $r[t_1,t_2] = \dfrac{Q(t_2)-Q(t_1)}{Q(t_1)}$

$Q(t)$: price index of time t.

ex: $t=0$ $1\$ \to u$, price per unit $= \frac{1}{u}$

$i=8\%$, $r=5\%$, p.p.u $t=1$?

$\$$: $1(1+0.08)^1 = 1.08$

p.p.u: $\frac{1}{u}(1+5\%) = \frac{1.05}{u}$

you can purchase $\dfrac{1.08}{\frac{1.05}{u}} = \dfrac{\overset{\to 1+i}{1.08}}{\underset{\to 1+r}{1.05}} u = \underbrace{1.02857}_{1+j} u$

$\Rightarrow$ 2.857 % more unit than last year

Inflation adjusted interest rate: (j)

$$1+j = \frac{1+i}{1+r}$$

$$\Rightarrow j = \frac{i+1}{1+r} - 1$$

Chapter 3.

Annuity : series of payment

Payment Period : time between 2 payments

term : time of 1st payment ~ time of last payment.

$\left\{\begin{array}{l}\text{level annuity : same payment amount} \\ \text{non-level annuity : vice versa}\end{array}\right.$

$\left\{\begin{array}{l}\text{annuity due : paid in advance} \\ \text{annuity immediate : paid in end.}\end{array}\right.$

$\left\{\begin{array}{l}\text{*annuity certain : has a term that is fixed.} \\ \text{contingent annuity : term depends on uncertain events.}\end{array}\right.$

Finite Sum : $\sum_{k=0}^{n-1} ar^k = a + ar + ar^2 \ldots + ar^{n-1} = \frac{a(1-r^n)}{1-r}$

Infinite Sum : $\sum_{k=0}^{\infty} ar^k = \sim \quad \sim = \frac{a}{1-r}$

$a_{\overline{n}|j}$: PV of an annuity that $= \frac{1-v^n}{j}$

- 1\$ per payment period.
- $n \Rightarrow$ # payments
- j effective interest per payment period.

$v = \frac{1}{1+j}$

- If payment = \$R $\Rightarrow$ PV = R $\cdot$ $a_{\overline{n}|j}$

$\ddot{a}_{\overline{n}|j}$: PV of annuity that $= \frac{1-v^n}{d} \to 1-v$

- 1\$ per payment period, Payable in advance.
- $n \Rightarrow$ # payments
- j effective interest per payment period.

where $v = (1+j)^{-1}$, d = effect discount rate

v : discount factor.

- If payment = \$R $\Rightarrow$ PV = $R \cdot \ddot{a}_{\overline{n}|j}$

$$\ddot{a}_{\overline{n}|j} = a_{\overline{n}|j} \cdot (1+j)$$

Proof: $\ddot{a}_{\overline{n}|j} = 1 + v + \ldots + v^{n-1}$

$= (v + v^2 + \ldots + v^n)(1+j) \to v = \frac{1}{(1+j)}$

$= a_{\overline{n}|j} \cdot (1+j)$

$$\ddot{a}_{\overline{n}|j} = 1 + a_{\overline{n-1}|j}$$

$s_{\overline{n}|j}$: AV at the end of last payment annuity.

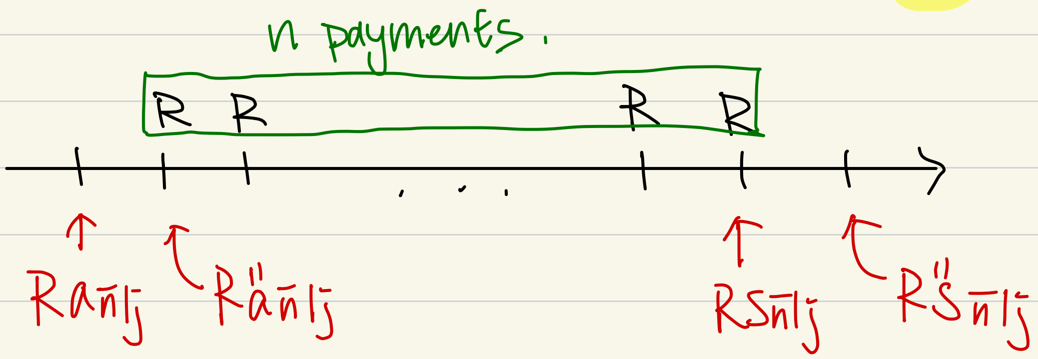
- Pays 1\$ per payment period Payable at end.
- n # payments.
- j = effective interest rate.

$$\Rightarrow \frac{(1+j)^n-1}{j}$$

$\ddot{s}_{\overline{n}|j}$: AV at the end of last payment annuity.

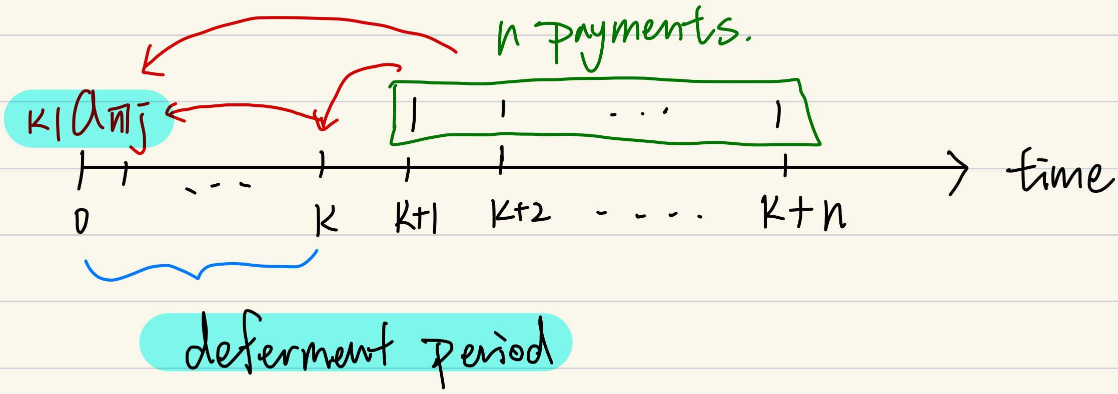
- Pays 1\$ per payment period Payable in Advance.
- n # payments.
- j = effective interest rate.

$$\Rightarrow \frac{(1+j)^n-1}{d}$$



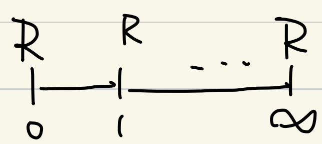
$$\begin{cases} \ddot{s}_{\overline{n}|j} = s_{\overline{n}|j}\cdot(1+j) \\ \ddot{s}_{\overline{n}|} = s_{\overline{n+1}|}-1 \end{cases}$$

$$\begin{cases} s_{\overline{n}|j} = a_{\overline{n}|j}\cdot(1+j)^n \\ \ddot{s}_{\overline{n}|j} = \ddot{a}_{\overline{n}|j}\cdot(1+j)^n \end{cases}$$



$$\begin{cases} {}_{k|}a_{\overline{n}|j} = a_{\overline{n}|j} \cdot v^{k} \\ {}_{k|}a_{\overline{n}|j} = \ddot{a}_{\overline{n+k+1}|j} - \ddot{a}_{\overline{k+1}|j} \end{cases}$$

Simple ordinary perpetuity $\Rightarrow$ forever.



$$PV = \frac{A}{i}$$

Chapter 4.

Suppose:

- $\bar{i}$ = effective interest rate per conversion period (1yr)
- m = # payments in one conversion period.
- n = term of annuity in conversion period.

PV of annuity with payment amount $\$\frac{1}{m}$:

$$a_{\overline{n}|\bar{i}}^{(m)} = \frac{1-v^n}{i^{(m)}}$$ (payment made at end)

$$\ddot{a}_{\overline{n}|\bar{i}}^{(m)} = \frac{1-v^n}{d^{(m)}}$$ (payment made at start)

FV of annuity ... $\$\frac{1}{m}$:

$$s_{\overline{n}|\bar{i}}^{(m)} = \frac{(1+\bar{i})^n-1}{\bar{i}^{(m)}}$$

$$\bar{i}^{(m)} = \left[(1+\bar{i})^{\frac{1}{m}}-1\right]m$$

$$\ddot{s}_{\overline{n}|\bar{i}}^{(m)} = \frac{(1+\bar{i})^n-1}{d^{(m)}}$$

Relationships:

$$\begin{cases} a_{\overline{n}|\bar{i}}^{(m)} = a_{\overline{n}|\bar{i}}\left(\frac{\bar{i}}{\bar{i}^{(m)}}\right) \\ \ddot{a}_{\overline{n}|\bar{i}}^{(m)} = \ddot{a}_{\overline{n}|\bar{i}}\left(\frac{d}{d^{(m)}}\right) \end{cases}$$

$$\begin{cases} s_{\overline{n}|\bar{i}}^{(m)} = s_{\overline{n}|\bar{i}}\left(\frac{\bar{i}}{\bar{i}^{(m)}}\right) \\ \ddot{s}_{\overline{n}|\bar{i}}^{(m)} = \ddot{s}_{\overline{n}|\bar{i}}\left(\frac{d}{d^{(m)}}\right) \end{cases}$$

Continuous annuities:

- $f(t)$ = rate of payment at time t.
- $a(t) = 1/v(t)$, is accumulation function.
- n = term of annuity.

$$\begin{cases} PV = \int_0^n v(t) f(t)\, dt \\ AV = PV \cdot a(n) = a(n) \int_0^n v(t) f(t)\, dt \end{cases}$$

Assume compound interest, with 1$ paid continuously over 1 year for total of n years.
such that:
- $f(t) = 1$
- $a(t) = e^{\delta t}$

☆
$$\begin{cases} PV = \bar{a}_{\overline{n}|} = \dfrac{1-v^n}{\delta} \\ AV = \bar{s}_{\overline{n}|} = \dfrac{(1+i)^n - 1}{\delta} \end{cases} \quad \text{where } v = e^{-\delta} = (1+i)^{-1}$$

Relationships:

$$\begin{cases} \bar{a}_{\overline{n}|} = a_{\overline{n}|}\left(\dfrac{i}{\delta}\right) = \ddot{a}_{\overline{n}|}\left(\dfrac{d}{\delta}\right) \\ \bar{s}_{\overline{n}|} = s_{\overline{n}|}\left(\dfrac{i}{\delta}\right) = \ddot{s}_{\overline{n}|}\left(\dfrac{d}{\delta}\right) \end{cases}$$

$$\begin{cases} \bar{a}_{\overline{n}|} = \lim\limits_{m\to\infty} a^{(m)}_{\overline{n}|} = \lim\limits_{m\to\infty} \ddot{a}^{(m)}_{\overline{n}|} \\ \bar{s}_{\overline{n}|} = \lim\limits_{m\to\infty} s^{(m)}_{\overline{n}|} = \lim\limits_{m\to\infty} \ddot{s}^{(m)}_{\overline{n}|} \end{cases} \quad \left(\text{Since } \lim\limits_{m\to\infty} i^{(m)} = \lim\limits_{m\to\infty} d^{(m)} = \delta\right)$$

↳ Force = nominal compounded continuously.

Arithmetic Progression: payment increase in constant amount.

- n payments
- first payment size P, increase by Q after
- $v = (1+j)^{-1}$

Increasing annuities

end of each period:

$$\begin{cases} PV = P a_{\overline{n}|j} + Q\left(\dfrac{a_{\overline{n}|j} - nv^n}{j}\right) \\ AV = P s_{\overline{n}|j} + Q\left(\dfrac{s_{\overline{n}|j} - n}{j}\right) \end{cases}$$

start of each period:

$$\begin{cases} PV = P \ddot{a}_{\overline{n}|j} + Q\left(\dfrac{a_{\overline{n}|j} - nv^n}{d}\right) \\ AV = P \ddot{s}_{\overline{n}|j} + Q\left(\dfrac{s_{\overline{n}|j} - n}{d}\right) \end{cases}$$

Special Case: $P = Q = 1$

$$\begin{cases} PV\ (\text{end}) = (Ia)_{\overline{n}|j} := \dfrac{\ddot{a}_{\overline{n}|j} - nv^n}{j} \\ AV\ (\text{end}) = (Is)_{\overline{n}|j} = \dfrac{\ddot{s}_{\overline{n}|j} - n}{j} \end{cases}$$

$$\begin{cases} PV\ (\text{start}) = (I\ddot{a})_{\overline{n}|j} := \dfrac{\ddot{a}_{\overline{n}|j} - nv^n}{d} \\ AV\ (\text{end}) = (I\ddot{s})_{\overline{n}|j} = \dfrac{\ddot{s}_{\overline{n}|j} - n}{d} \end{cases}$$

Decreasing annuities

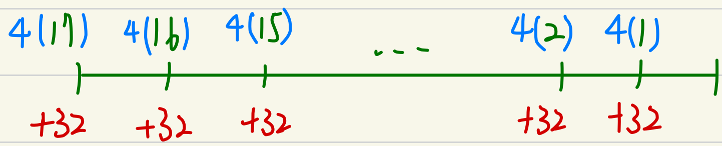
Special case : $P = n \; , \; Q = -1$ *

$$\begin{cases} PV\,(\text{end}) = (Da)_{\overline{n}|j} = \dfrac{n - a_{\overline{n}|j}}{j} \\ AV\,(\text{end}) = (Ds)_{\overline{n}|j} = \dfrac{n(1+j)^n - s_{\overline{n}|j}}{j} \end{cases}$$

$$\begin{cases} PV\,(\text{start}) = (D\ddot{a})_{\overline{n}|j} = \dfrac{n - a_{\overline{n}|j}}{d} \\ AV\,(\text{start}) = (D\ddot{s})_{\overline{n}|j} = \dfrac{n(1+j)^n - s_{\overline{n}|j}}{d} \end{cases}$$

ex : PV of annuity due , 17 payments of 100, 96, 92, ... 36

$100 \xrightarrow{-4} 96 \xrightarrow{-4} 92$



$Q = -1 \; , \; n = 17$

$$PV = 4\,(D\ddot{a})_{\overline{17}|j} + 32\,\ddot{a}_{\overline{17}|j}$$

$$= 4\left[\frac{17 - a_{\overline{17}|j}}{d}\right] + 32\,\ddot{a}_{\overline{17}|j}$$

Perpetuity : annuity with no end date

$$\begin{cases} PV\,(\text{end}) = a_{\overline{\infty}|j} = \lim\limits_{n\to\infty} \dfrac{1-(1+j)^{-n}}{j} = \dfrac{1}{j} \\ PV\,(\text{start}) = \ddot{a}_{\overline{\infty}|j} = \lim\limits_{n\to\infty} \dfrac{1-(1+j)^{-n}}{d} = \dfrac{1}{d} \end{cases}$$

Increasing Perpetuity:

$$PV(\text{end}) = (Ia)_{\overline{\infty}|j} = \frac{1}{j} + \frac{1}{j^2}$$

$$PV(\text{start}) = (I\ddot{a})_{\overline{\infty}|} = \frac{1}{d^2}$$

Chapter 5

Amortization Schedule:

$$\begin{cases} L : \text{loan amount} \\ n : \text{pmt periods} \\ R : \text{pmt amount} \end{cases}$$

$$(\Sigma)\begin{cases} B_k : \text{loan balance after kth payment} \\ I_k : \text{interest of kth payment} \\ P_k : \text{principal of kth payment} \end{cases}$$

$I_k = B_{k-1} \cdot i$

$P_k = R - I_k$

$B_k = B_{k-1} - P_k$

$B_k = L(1+i)^k - P s_{\overline{k}|i}$

$= R a_{\overline{n-k}|i}$

$B_k = L(1+i)^k$ − AV of pmt paid at time t

B_k = PV of remaing pmt

Special case : (fully paid off by level pmts)

$\Rightarrow L = R a_{\overline{n}|i}$

PV of L at (k-1)

$I_k = B_{k-1} \cdot i = (R a_{\overline{n-(k-1)}|}) \cdot i = R \dfrac{(1 - v_i^{n-k+1})}{\cancel{i}} \cdot \cancel{i}$

$P_k = R - I_k = R - R(1 - v_i^{n-k+1}) = R v_i^{n-k+1}$

$\Sigma P_k = \Sigma R v_i^{n-k+1} = R(v^n + v^{n-1} + \cdots v)$

$\Sigma I_k = \Sigma(R - P_k) = nR - \Sigma P_k = nR - L$ $(\Sigma P_k = L)$

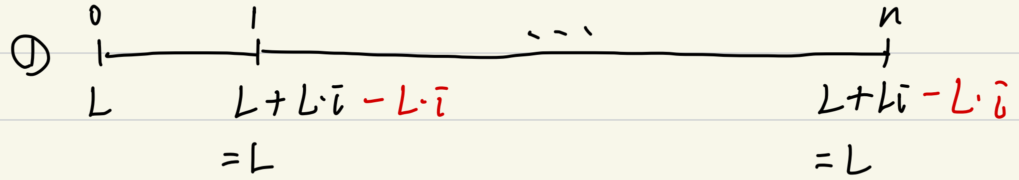
$B_k = L(1+i)^k - R s_{\overline{k}|i}$

$B_k = R a_{\overline{n-k}|i}$ Loan balance

$* \dfrac{P_{k+1}}{P_k} = 1 + i$

Sinking fund:

- an interest stream & a fund to accumulate L



L : loan amount
i : interest on loan
j : interest on sinking fund
D : PV of L

$I_k = L \cdot i$

pmt each period $= Li + D$ $\left(\frac{L}{s_{\overline{n}|}}\right)$

book value of debt after kth pmt $= L - D\, s_{\overline{k}|j}$

↳ basically outstanding balance.

net interest pmt $= Li - (D\, s_{\overline{k-1}|j}) \cdot j$

charged − earned.

AM vs. SFM

If $i = j$, same

$i > j$, $R_{AM} < R_{SFM}$

$i < j$, $R_{AM} > R_{SFM}$

Chapter 6

Bonds:

- F: face value
- N: terms (yr)
- m: coupon frequency ($m = 2$)
- n: number of coupons ($N \cdot m$)
- α: nominal annual coupon rate compounded m/yr

Coupon amount: $F \cdot r = F \cdot \alpha / m$

- C: redemption amount
- $C = F \Rightarrow$ redeemed at par
  else, par value bond.

$$P = Fr \cdot a_{\overline{n|}} + C \cdot v^n \quad \text{(price)}$$
$$= C + (Fr - Cj)\, a_{\overline{n|}}$$

B_t = Book value $= Fr\, a_{\overline{n-t|}} + C\, v^{n-t}$ → PV of remaining pmt.

$B_0 = P$

$B_n = C$

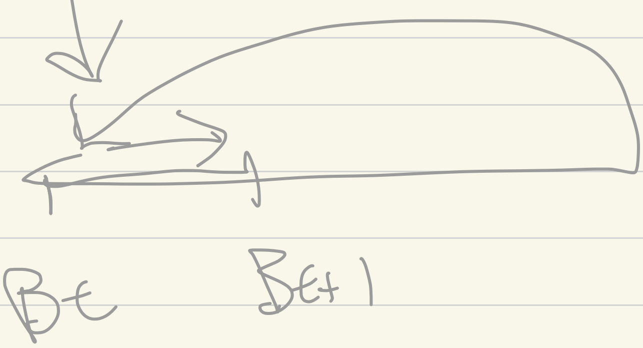
think abt amortization

$I_t = B_{t-1} \cdot j$

$P_t = Fr - I_t$ (Book value paid by tth coupon)

$Fr = I_t + P_t$

$$\frac{P_{t+1}}{P_t} = 1 + j$$

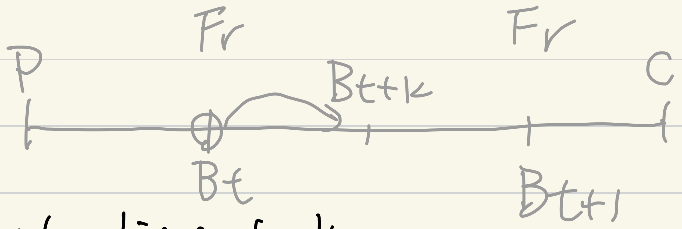


$B_{t+1} = B_t - P_{t+1}$ → Book value immediately after tth coupon paid. ($-$ principal paid)

$= B_t - (Fr - I_{t+1})$

$= B_t - (Fr - B_t \cdot j)$

$= B_t(1+j) - Fr$



(Dirty price) (full)

B_{t+k} = PV of remaining pmt. at time $t+k$

$= B_t \cdot (1+j)^k$

(market)

Clean Price (quotation) $Q = B_t(1+j)^k - kFr$ → remove portion earned by seller.

$k \Rightarrow$ find with actual/actual

accrued interest : $I = B_{t+k} - Q = kFr$

↳ seller's earned portion of the $(t+1)$th coupon

Callable bonds : bonds where issuer have right to redeem before maturity.

call dates : dates issuer can redeem

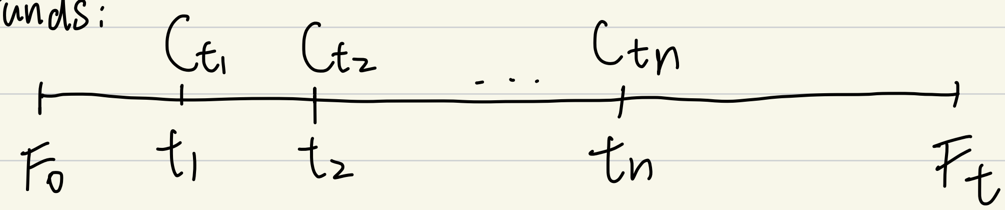
lockout period : period before first call date.

call premium : $C^* - C$

Zero coupon bonds : $P = Cv^n$

Chapter 7

Funds:



C = contribution

$$F_T = F_0(1+\bar{i})^T + \sum_{j=1}^{n} C_{t_j}(1+\bar{i})^{T-t_j}$$

↳ $F_T(1+\bar{i})^{-T} = F_0 + \Sigma C_{t_j}(1+\bar{i})^{-t_j}$

$\bar{i}$ = yield rate = DWYR or IRR

$$F_1 = F_0 + \Sigma C_{t_j} + I.$$

$$\Rightarrow \bar{i} = \frac{I}{F_0 + \Sigma C_{t_j}(1-t_j)}$$

$F_{t_j^-}$ = fund amount <u>before</u> contribution C_{t_j}

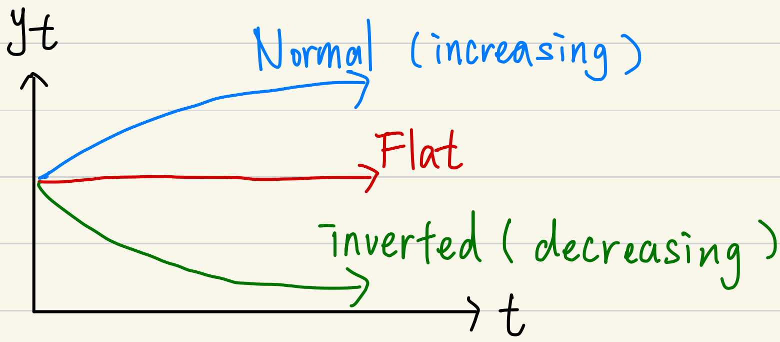
$F_{t_j^+}$ = fund amount <u>after</u> contribution C_{t_j}

$$1+\bar{i}_k = \frac{F_{t_k^-}}{F_{t_{k-1}^+}}$$

$$\Rightarrow (1+\bar{i})^T = (1+\bar{i}_1)(1+\bar{i}_2)\cdots(1+\bar{i}_{n+1})$$

Chapter 8

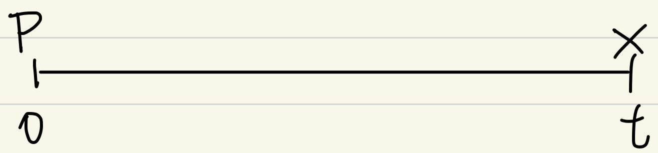
Yield curve:



Spot rates : set of yield rates.

→ a.k.a zero coupon rate

(S_t) r_t = annual i earned by investment at $t=0$ having a single pmt at maturity date t



$\Rightarrow$ Price $= X(1+r_t)^{-t}$

y_t = annual i earned by an investment at $t=0$ having pmts that can occur throughout $[0,t]$

ex: bonds $P = Fr\, a_{\overline{n}|y_n} + C(1+y_n)^{-n}$

if zero coupon, $P = C(1+y_t)^{-t} = C(1+r_t)^{-t}$

bootstrapping: deduce spot rates

theoretical **foward rate** :

$({}_s\overline{s}_{t-s})$ $f_{[s,t]}$ = annual effective interest rate over $[s,t]$

$$(1+f_{[s,t]})^{t-s} = \frac{(1+r_t)^t}{(1+r_s)^s}$$

if compounded $\left(1+\frac{f_{[s,t]}}{m}\right)^{m(t-s)} = \frac{\left(1+\frac{r_t}{m}\right)^{mt}}{\left(1+\frac{r_s}{m}\right)^{ms}}$

Chapter 9

Price Curve:

PV / price of cash flow at interest i

$$P(i) = \Sigma C_{t_k} V_i^{t_k}$$

assumption unless specified.

Macaulay duration : $D(i,\infty) = -\frac{1}{P(i)} \cdot \frac{dP}{d\delta}$

Modified duration : $D(i,m) = -\frac{1}{P(i)} \cdot \frac{dP}{di^{(m)}}$

negative relative roc in price when i changes.

$$D(i,\infty) = \frac{\sum_{k\geq 0} t_k \cdot C_{t_k} V_i^{t_k}}{P(i)}$$

weight.

$$= \sum_{k\geq 0} \left(\frac{C_{t_k} V_i^{t_k}}{P(i)} \right) t_k .$$

$$D(i,m) = D(i,\infty) \cdot (V_i)^{\frac{1}{m}}$$

$$= \frac{\sum_{k\geq 0} t_k C_{t_k} V_i^{t_k + \frac{1}{m}}}{P(i)}$$

n-year zero-coupon bond w/ C : $D(i,\infty) = n$

n-year bond annual coupons Fr & C :

$$D(i,\infty) = \frac{Fr \cdot (Ia)_{\overline{n}|i} + n \cdot C v_i^n}{Fr \cdot a_{\overline{n}|} + C \cdot v_i^n}$$

n-year annuity-immediate R : $D(i,\infty) = \frac{(Ia)_{\overline{n}|i}}{a_{\overline{n}|i}}$

Convexity : measure of curvature b/w $P(\bar{i})$ and $\bar{i}$

$$C(i,\infty) = \frac{1}{P(i)} \cdot \frac{d^2P}{d\delta^2} \qquad (\delta = \ln(1+i))$$

$$C(\bar{i},m) = \frac{1}{P(\bar{i})} \cdot \frac{d^2P}{d\bar{i}^{(m)2}}$$

$$C(i,\infty) = \frac{\sum_k t_k^2 \cdot C_{t_k} V_i^{t_k}}{P(i)} = \sum_{k\geq 0} \left(\frac{C_{t_k} V_i^{t_k}}{P(i)} \right) t_k^2$$

$$C(\bar{i},m) = \frac{C(\bar{i},\infty) + \frac{1}{m} D(\bar{i},\infty)}{(1+\bar{i})^{\frac{2}{m}}}$$

$$P(i)\Big|_{\bar{i}^{(m)} = \bar{i}^{(m)}_{new}} - P(i)\Big|_{\bar{i}^{(m)} = \bar{i}^{(m)}_{old}}$$

$$= P(i)\Big|_{\bar{i}^{(m)} = \bar{i}^{(m)}_{old}} \left[-\left(D(\bar{i},m)\Big|_{\bar{i}^{(m)} = \bar{i}^{(m)}_{old}}\right)\Delta + \frac{\left(C(\bar{i},m)\Big|_{\bar{i}^{(m)} = \bar{i}^{(m)}_{old}}\right)\Delta^2}{2!} \right]$$

where $\Delta = \bar{i}^{(m)}_{new} - \bar{i}^{(m)}_{old}$

$A_t \geq 0$: total inflow

$L_t \geq 0$: total outflow

net cashflow : $A_t - L_t$

Surplus $S(\bar{i}) = \sum_{t\geq 0} (A_t - L_t)(1+\bar{i})^{-t}$

$$= \sum_{t\geq 0} A_t (1+\bar{i})^{-t} - \sum_{t\geq 0} L_t (1+\bar{i})^{-t}$$

Redington immunization :

→ positive surplus maintained a small fluctuation.

Conditions :

① $P_A(\bar{i}_0) = P_L(\bar{i}_0)$

② $D_A(\bar{i}_0, \infty) = D_L(\bar{i}_0, \infty)$

③ $C_A(\bar{i}_0, \infty) \geq C_L(\bar{i}_0, \infty)$